cost of 3 Kg of Wheat, 2 Kg of Sugar and 1 Kg of rice is ₹170. Find the cost of per kg each item using matrix inversion method.

10. The data are about an economy of two industries A and B. The values are in crores of rupees.

Producer	User		Final demand	Total output
	A	B		
A	50	75	75	200
B	100	50	50	200

Find the output when the final demand changes to 300 for A and 600 for B

Summary

- Determinant is a number associated to a square matrix.
- If $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ then its $|A| = a_{11}\,a_{22} - a_{21}\,a_{12}$
- Minor of an arbitrary element a_{ij} of the determinant of the matrix is the determinant obtained by deleting the ith row and j^{th} column in which the element a_{ij} stands. The minor of a_{ij} is denoted by M_{ij}.
- The cofactor is a signed minor. The cofactor of a_{ij} is denoted by A_{ij} and is defined as $A_{ij} = (-1)^{i+j} M_{ij}$.
- Let $\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$, then

  $\Delta = a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13}$ or $a_{11}M_{11} - a_{12}M_{12} + a_{13}M_{13}$
- adj A $= [A_{ij}]^T$ where $[A_{ij}]$ is the cofactor matrix of the given matrix.
- $|adjA| = |A|^{n-1}$, where n is the order of the matrix A.
- $A(adjA) = (adjA)(A) = |A|I$.
- $adjI = I$, I is the unit Matrix.
- adj(AB) = $(adj\ B)(adj\ A)$.
- A square matrix A is said to be singular, if $|A| = 0$.
- A square matrix A is said to be non-singular if $|A| \neq 0$
- Let A be any square matrix of order n and I be the identity matrix of order n. If there exists a square matrix B of order n such that $AB = BA = I$ then, B is called the inverse of A and is denoted by A^{-1}
- Inverse of A (if it exists) is $A^{-1} = \frac{1}{|A|} adjA$.
- Hawkins - Simon conditions ensure the viability of the system. If B is the technology matrix then Hawkins-Simon conditions are

  (i) the main diagonal elements in $I - B$ must be positive and

  (ii) $|I - B|$ must be positive.

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